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Trimethyl Borate as Film-forming Electrolyte Additive to Improve High-voltage Performances

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Abstract

Enhancing the stability of the interface between the electrode and electrolyte at high voltages is crucial concerning the development of Li-ion batteries with high energy densities. Application of some additives in the electrolyte is not only the simplest but also the most effective way to form a protection layer against the electrolyte decomposition and the electrolyte corrosion to the electrode. Herein, we introduce trimethyl borate (TMB) as an additive of the commercial electrolyte to ameliorate the performance of a LiCoO₂ cell charged to 4.5 V because its addition lowers the oxidation potential of the baseline electrolyte (3.75 V vs. 4.25 V). By being oxidized preferentially and thus forming a compact protection layer of about 25 nm thick on the cathode surface, the additive suppresses the electrolyte decomposition and protect the LiCoO₂ cathode against the structural degradation. The capacity retention of the cell after 100 cycles between 2.5 V and 4.5 V at 0.1 C increases from 64% to 81% when 2.0 wt.% TMB is added into the baseline electrolyte. The X-ray photoelectron spectroscopic results demonstrate the oxidation of TMB on the cathode and therefore, the suppressed decomposition of the electrolyte. The results of the X-ray diffraction and Raman spectroscopy show the better structural maintenance of the LiCoO₂ material in the TMB-containing electrolyte. The protection mechanism of the TMB additive was comprehensively studied.

Keywords FILM-FORMING; ADDITIVE; BORON-CONTAINING; HIGH-VOLTAGE; LI-ION BATTERY

Introduction

With severe environmental concerns emerging due to the excess consumption of fossil fuels, people are paying more attention to the research and evolution of new sources of green energy. As a result, the demands for batteries with higher energy density are becoming more urgent than ever before.^[1] Raising the cut-off charge voltage of the cathode material can increase both the specific capacity and the average working voltage of the material and, therefore, leads to the increase of the energy density of the battery. However, severe electrolyte decomposition often occurs on the cathode surface and harm the structure of the cathode material at high voltages. Furthermore, the electrolyte decomposition products will make up a thick layer on the cathode, increasing the interface resistance and the cell polarization. All these factors will have a bad influence on the cycling performance of the battery.^[2] Doping of elements^[3] was studied to keep the structure of cathode material stable at high voltages. However, it is not easy to obtain a uniformly doped material, especially when the material is synthesized by solid-state reaction and the doping content is low. Making the cathode material surface coated with a layer of inert species (e.g. Al₂O₃^[4]) is another strategy to lower the activity of the surface and to protect its bulk. However, it is not easy to form a compact coating layer on the surface.^[5] To form a protection film on the cathode material

by adding a film-forming additive into the electrolyte seems to be a practical and simple method to improve the cycling stability of the battery.

To improve the cycling performance of the battery at high voltages, many sorts of additives have been studied such as phosphate,^[6] sulfonate,^[7] nitrile^[8] and borate ester.^[9] Boron-containing compounds are one of the most attractive additives.^[10] It was reported that the addition of trimethyl boroxine can effectively improve the cycling performance of $\text{Li}_{1.2}\text{Mn}_{0.5}\text{Ni}_{0.13}\text{Co}_{0.12}\text{O}_2$ (LNCM) at 4.5 V.^[11] However, trimethyl boroxine can subsequently catalyze the oxidation of the carbonate-based electrolytes.^[12] In addition, the oxidation products of trimethyl boroxine can further react with the hydrolysis products of LiPF_6 (HF and POF_3 , for example), forming chemicals harmful to both the cathode material and the battery system.^[13] Adding 10.0 wt.% triethyl borate can also enhance the cycling performance of LNCM without accelerating the following oxidation of the electrolyte.^[9] However, the content of triethyl borate was too high to be applied in the Li-ion batteries. As far as we know, not many studies have been conducted to investigate the boron-containing additives on the electrochemical performance of LiCoO_2 above 4.5 V. Boron-containing additives such as lithium bis(oxalate)borate (LiBOB)^[14], lithium difluoro(oxalato)borate (LiDFOB)^[15] and tris(trimethylsilyl) borate (TMSB)^[16] can be oxidized on the cathode preferentially during the charge process due to their higher highest occupied molecular orbital (HOMO) energies. Additionally, it was reported that, because of the electron deficiency of boron, the borate compounds can prevent the PF_6^- anion from decomposition into PF_5 , LiF or other by-products.^[17] The electrophilic boron-containing compounds can also increase the solubility of LiF, which would otherwise make the interface impedance increase. Moreover, the borate compounds can be regarded as anion receptors; they complex with the anions in the lithium salt. As a result, not only the conductivity but also the transference number of the Li-ions in the electrolyte are enhanced^[18-19].

In this work, trimethyl borate (TMB) was proposed to be a film-forming additive as it has a molecular structure similar to that of trimethyl boroxine. Both of these two molecules contain the B-O bond, but TMB might possess lower oxidization potential than the ring-structured trimethyl boroxine. The oxidation potential of the trimethyl boroxine on the LiCoO_2 cathode in the cell is 3.85 V^[20] while that of TMB is 3.75 V. The operation mechanism of TMB is studied through electrochemical evaluation and physical characterization. TMB was tested as an electrolyte additive on LNCM^[21] and layered lithium-rich oxide (LLO)^[22] cathode at 0.5 C without much further discussion concerning the mechanism before. In addition, a lower cycling rate (0.1 C) was applied in this work in order to have more Li-ions extracted from the LiCoO_2 and therefore reach a stronger oxidizing capability to the electrolyte. As a consequence, the requirements for the cycling/oxidation stability of the electrolyte are stricter. Additionally, lower current density allows the decomposition process of both the additive and the baseline electrolyte more complete and the following characterization results more apparent and reliable. Electrochemical evaluation shows that an addition of 2.0 wt.% TMB into the carbonate electrolyte can effectively enhance the cycling performance of the LiCoO_2 charged to 4.5 V at 0.1 C; its capacity retention after 100 cycles increases from 64% in the baseline electrolyte to 81% in the TMB-containing electrolyte.

Experimental

Commercial 1.0 mol L^{-1} LiPF_6 EC/DMC (1:1 v/v; EC for ethylene carbonate, DMC for dimethyl carbonate) was purchased from BASF Chemical Co. Ltd (China) and used as the baseline electrolyte. Trimethyl borate (TMB; 99.9995+%) was a product of Alfa Aesar Chemical Co. Ltd (China). Lithium cobalt oxide (LiCoO_2) was a product of Beijing Easpring Material Technology Co. Ltd (China). The electrolytes containing 1 wt.%, 2 wt.%, and 3 wt.% TMB were prepared in an argon-filled glove box

(MBraun Lab Master 130), in which the contents of the oxygen and water were controlled below 0.1 ppm. Coin cells (CR2032) were assembled in the glove-box with metallic Li foil as the counter electrode, an Al sheet coated with a slurry containing of 80 wt.% LiCoO₂ powder, 10 wt. % carbon nanotube and 10 wt.% PVDF binder as the working electrode.

The cycling performance of the Li/LiCoO₂ coin cell was galvanostatically cycled between 2.5 and 4.5 V (1 C = 200 mAh g⁻¹) on the Land CT2001A battery tester (Wuhan, China). The linear sweep voltammetry (LSV) and cyclic voltammetry (CV) were recorded on the electrochemical workstation (CH Instruments). The electrochemical impedance spectroscopy (EIS) was conducted over a frequency range from 4 MHz to 1 mHz on the electrochemical workstation (Zahner IM6, Germany).

Scanning electron microscope (SEM, HITACHI S-4800) and transmission electron microscope (TEM, JEM2100plus) were used to observe the surface morphology of the LiCoO₂ electrode. X-ray photoelectron spectroscopy (XPS, Thermo Fisher ESCALAB 250 Xi, Thermo Fisher with monochromatic 150W Al K α radiation) was used to analyze the chemical state of the element in the SEI layer. Fourier-transformed infrared spectroscopy (FTIR, a Bruker VERTEX 70 V) was used for detecting the chemical bonds on the LiCoO₂ electrode surface. A LabRAM HR Evolution Raman spectrometer and an X-ray diffractometer (XRD, X'Pert-Pro MPD with monochromatic Cu K α radiation λ =0.1541 nm) were applied to characterize the structural variation of the LiCoO₂ cathodes before and after 100 cycles. The cells were charged potentiostatically at 4.5 V for 10 h before disassembled for the FTIR characterization. In terms of the other physical characterizations, the cells were tested at 0.1 C between 2.5 V and 4.5 V for 100 cycles. Before the above physical characterization, the electrochemically cycled coin cells were disassembled in the glove box. The LiCoO₂ electrode was rinsed in pure DMC for three times to remove the LiPF₆ salt and the solvent.

Gaussian 09 package was applied to perform the calculations. The geometric structures of the molecule/complex were optimized with B3LYP method at 6-311++ G (2d,2p) basis set. The solvent effect was considered by introducing a polarized continuum model using a permittivity of 45.

Results and Discussion

(1) Reduced interface reactivity

An additive that can form an SEI film is supposed to own a high oxidation activity so that it can oxidize on the cathode preferentially, in order to generate a film that can prevent the oxidation of the baseline electrolyte and protect the cathode materials. Therefore, it is important for an effective film-forming additive to have a high oxidation activity.^[5] The calculated HOMO energy of TMB (-0.29 eV) is higher than that of the carbonate solvents (*e.g.*, -0.32 eV for EC), indicating that losing one electron is easier for TMB than for EC. This will result in a lower onset oxidation potential for TMB than for the carbonate solvents.

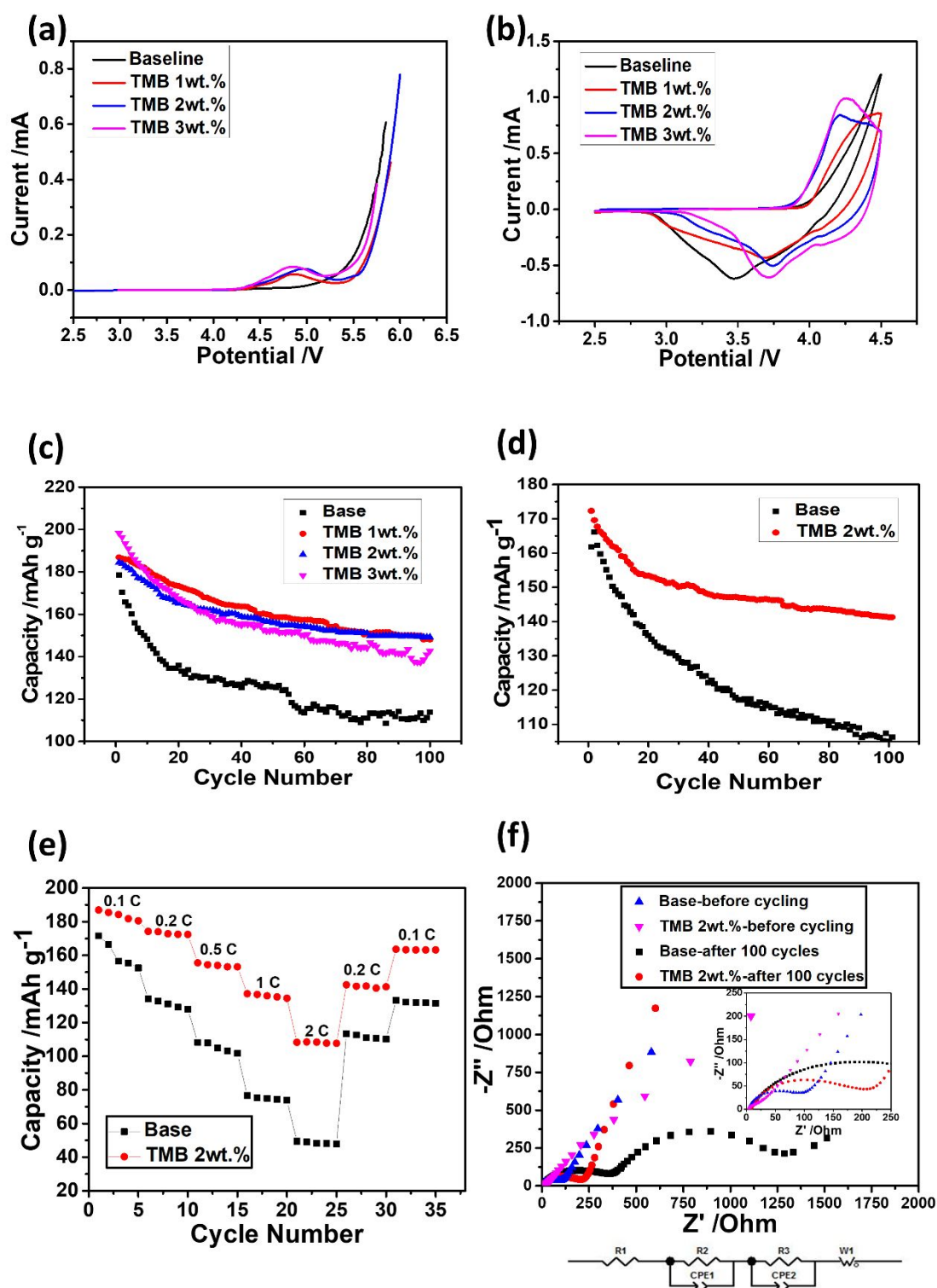


Figure 1. The LSV profiles of the baseline and TMB-containing electrolytes on a Pt electrode at 0.1 mV s⁻¹ from the open circuit voltage to 6.0 V (a), the CV curves of the Li/LiCoO₂ cells with baseline and TMB-containing electrolytes in the initial cycle at 0.5 mV s⁻¹ between 2.5 and 4.5 V (b), the cycling performance at 0.1 C (c), the cycling performance of the Li/LiCoO₂ cells with the baseline and

2.0 wt.% TMB-containing electrolytes at 1.0 C (d), the rate performance of the Li/LiCoO₂ cells with the baseline and 2.0 wt.% TMB-containing electrolytes (e), and the EIS spectra of the Li/LiCoO₂ cells using different electrolytes before and after 100 cycles and the equivalent circuit (f)

Figure 1a presents the LSV profiles of the electrolytes with and without TMB. It is seen that compared with the baseline electrolyte, the TMB-containing electrolytes are oxidized at a lower potential (4.25 V vs. 5.00 V). Figure 1b shows the initial CV curves of the electrolyte on the LiCoO₂ electrode. The onset oxidation potential of the TMB-containing electrolyte is lower in the charging curve, indicating that TMB is oxidized in advance. In the second cycle (Figure S1a), this difference disappears, revealing the completion of the TMB oxidation. It can be inferred from the CV curves that the onset oxidation potential of TMB in the Li/LiCoO₂ cell is around 3.75V, indicating the transition metal oxide LiCoO₂ has a catalysis effect on the oxidation of TMB, in comparison with the 4.25V onset oxidation potential of TMB in the LSV curve on the inert Pt electrode.

The onset oxidation potential of TMB is verified with the initial charge/discharge curves of the Li/LiCoO₂ cells in the baseline and TMB-containing electrolytes. In comparison with the 1st charge curve of the LiCoO₂ in the baseline electrolyte, the charge curve of the LiCoO₂ in the TMB-containing electrolyte shows an excess oxidation progress that begins at around 3.75 V (Figure S1b and S1c). Its intensity increases with the increasing TMB concentration and does not appear in the 2nd cycle. The cell using the baseline electrolyte shows an initial discharge capacity of 178.5 mAh g⁻¹, with a coulombic efficiency of 86.8%. The cells using electrolyte that contains 1.0 wt.%, 2.0 wt.% and 3.0 wt.% TMB deliver higher initial discharge capacities of 186.9, 184.4 and 198.3 mAh g⁻¹ with lower coulombic efficiencies of 79.1%, 74.5% and 70.3%, respectively. Three factors are supposed to be responsible for the relatively low initial discharge capacity of the cell using the electrolyte with the addition of 2.0 wt.% TMB, the conductivity of the electrolyte, the thickness of the SEI layer, and the reactivity between the TMB decomposition products and the lithium salt. The cathode in the 3.0 wt.% TMB-containing electrolyte has the thickest SEI film and the most severe reaction between the TMB decomposition products and the lithium salt^[23-24]. The cells with 1.0 wt.% TMB-adding electrolyte have the thinnest SEI film. The cell with 2.0 wt.% TMB-adding electrolyte reaches a balance among these factors and shows the best performance. The lower coulombic efficiency results from the irreversible oxidation of the TMB additive forming an SEI film as well as the irreversible phase transition of the LiCoO₂ cathode material. However, the SEI film formed by the TMB decomposition products prevents the further decomposition of the electrolyte and leads to low interface resistance (polarization suppressed) as shown in the following EIS results and higher coulombic efficiencies (Figure S1d). As a result, adding TMB into the electrolyte can be a way to improve the reversible capacity as well as the cycling performance of the cell.

(2) Improved cycling performance

Figure 1c shows the cycling performance of the Li/LiCoO₂ cells using electrolytes containing different contents of TMB at 0.1 C. The capacity retention of the Li/LiCoO₂ cell increases from 64% to 79%, 81% and 72% after 100 cycles, respectively, when 1.0, 2.0 and 3.0 wt.% TMB are added into the baseline electrolyte. Addition of too much TMB in the baseline electrolyte results in lower capacity retention because some of the TMB decomposition products can react with LiPF₆ and form lithium boron oxide (LBO)^[23-24]. Therefore, we chose the electrolyte that contains 2.0 wt.% TMB for further experiments in the following discussion.

Figure 1d shows the cycling performance of the Li/LiCoO₂ cells using the electrolyte with and without 2.0 wt.% TMB at 1.0 C. The cell using the electrolyte with 2.0 wt.% TMB addition shows an improved capacity retention of 82% after 100 cycles with a higher initial discharge capacity of 172.3

mAh g⁻¹. By contrast, the cell using the baseline electrolyte shows a capacity retention of only 65% after 100 cycles with an initial discharge capacity of 161.8 mAh g⁻¹. Decreasing the cycling rate from 1.0 C to 0.1 C does not lower the capacity retention of the cell. Considering that low current density allows more complete chemical reaction, we use the cells working at 0.1 C for the subsequent characterizations. In addition, the gap of the discharge capacity between the cells using electrolytes with and without TMB additive is larger at 1.0 C. This implies that the TMB-derived SEI film has a lower interphase resistance, as is confirmed with the following EIS test.

The rate performances of the LiCoO₂ electrode in the baseline electrolyte and the electrolyte with 2.0 wt.% TMB added are displayed in Figure 1e. It is obvious that the LiCoO₂ electrode in the 2.0 wt.% TMB-added electrolyte shows a better rate performance than that in the baseline electrolyte. This enhancement is due to the TMB-derived protective SEI film, which restrains the continuous decomposition of the electrolyte at high voltage and alleviate the over-growth of the SEI layer^[16]. Moreover, the difference of discharge capacity between the cells with and without TMB additive becomes larger with the increasing current density. This is mainly attributed to the surface film modified by TMB with a low interfacial resistance and weak polarization at a large current density, beneficial for facilitating the Li-ion migration and delivering a high discharge capacity^[25]. This outcome is in accordance with the results in Figure 1d.

Figure 1f shows the EIS spectrum of the Li/LiCoO₂ cells before and after 100 cycles. Each EIS spectrum contains two semicircles that are at high and medium frequencies respectively and a slope which is at low frequency. The semicircles located at high frequency and the medium frequency stand for the resistance of the SEI layer (R_{SEI}) and the charge transfer (R_{ct}), respectively, and the slope line reflects the Warburg impedance. However, when the capacitances of the double electrical layer and the film are close to each other, the two semicircles will merge into one. The equivalent circuit diagram is also shown in Figure 1f, in which R_1 refers to the inner resistance of the cell, R_2 for the R_{SEI} and R_3 for the R_{ct} . The corresponding fitting data is shown in Table S1. The results show the smaller R_{SEI} in the cell with the TMB-containing electrolyte than that of the cell using the baseline electrolyte, indicating that the contact between the electrolyte and electrode can be inhibited and the integrity of the cathode can be protected to some extent by the TMB-derived SEI during the cycling process^[26]. Larger R_{SEI} means that more baseline electrolyte is decomposed and a thicker SEI layer on the cathode. The cathode surface is covered with the decomposition product and the Li⁺ extraction resistance is increased at high voltages^[27]. As a result, the R_{ct} of the cell with TMB addition is smaller than that of the cell with the baseline electrolyte. This indicates that the LiCoO₂ cathode in the cell with TMB addition is more stable when charged to high potential, beneficial for improving the cycling stability of the cell. This explains the improved rate performance of the cell using the electrolyte with the addition of 2.0 wt.% TMB.

(3) Protective surface layer

The surface morphology of the cycled LiCoO₂ electrodes were characterized with the SEM and TEM techniques. XPS and FTIR were carried out to recognize the composition of the SEI film. The SEM imaging (Figure 2a, b, d and e) shows that a thick and heterogeneous film covers the surface of the LiCoO₂ from the cell with the baseline electrolyte after 100 cycles due to the continuous decomposition of the electrolyte. In contrast, the surface of the LiCoO₂ from the cell with the TMB-containing electrolyte is covered with a film that is thin and uniform of *ca.* 25 nm thick (Figure 2c and f). These suggest that TMB can help to hinder the continuous decomposition of the baseline electrolyte, beneficial for effectively protecting the LiCoO₂ electrode but without increasing its resistance significantly. The composition of the SEI film is analyzed below. With the XPS results, it is confirmed that TMB can be

oxidized in advance and help to hinder the continuous decomposition of the baseline electrolyte.

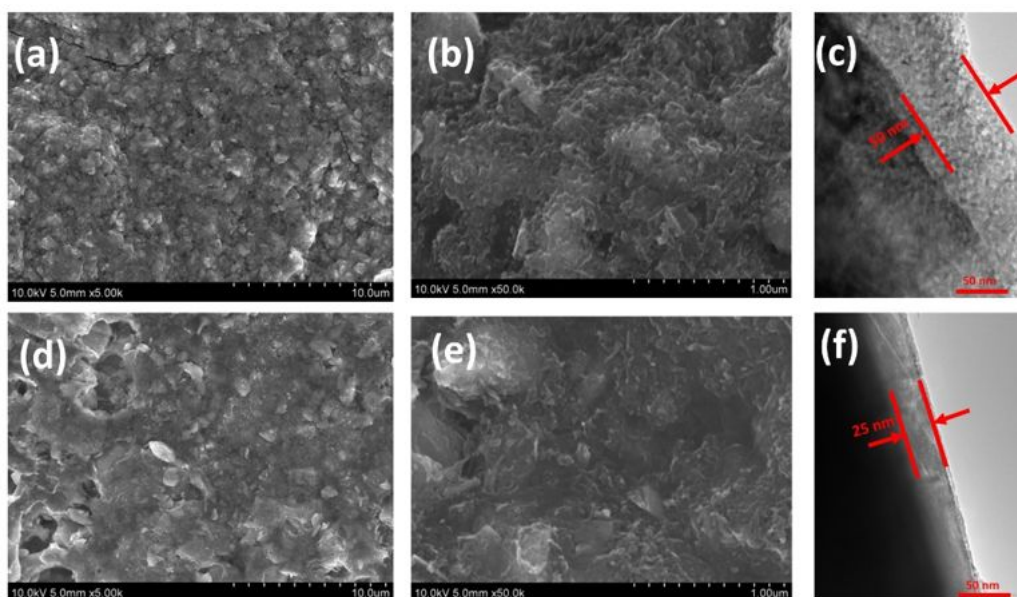


Figure 2. The SEM (a, b, d and e) and TEM (c and f) images of the LiCoO_2 electrodes after 100 cycles in baseline (a, b and c) and TMB-containing electrolyte (d, e and f)

The composition of the SEI layer on the LiCoO_2 electrodes cycled in the electrolytes with and without the addition of 2.0 wt.% TMB was analyzed with XPS (Figure 3). The four peaks in the C 1s spectrum at 284.8, 286.2, 289.8 and 291.2 eV correspond to the C-C bond, C-O bond, C=O bond and C-F bond,^[1] respectively. The C-C bond is mainly from the added conductive agent carbon black. The C-O bond and C=O bond are most likely related to the electrolyte decomposition products such as ROCO_2Li .^[28] The C-F bond comes from the PVDF. The weaker C=O peak and C-O peak in the LiCoO_2 electrode from the cell using the TMB-containing electrolyte reveals that less electrolyte is decomposed on the electrode, implying that the protection effect of the TMB oxidation products against the mass decomposition of the baseline electrolyte.

The O 1s spectrum of the LiCoO_2 electrode in the baseline electrolyte contains two peaks, the peak at 532.2 eV for the C-O bond and the peak at 534.1 eV for the C=O bond,^[29] which is consistent with the C 1s spectrum. It can be observed that two more peaks appear in the O 1s spectrum of the cycled LiCoO_2 electrode in the TMB-containing electrolyte, at 533.2 eV for the B-O bond^[30] and at 529.5 eV for the Li-O bond in the lattice of LiCoO_2 ^[31], respectively. As LiCoO_2 is covered with the SEI layer, the appearance of the 529.5 eV peak indicates that the SEI layer on the LiCoO_2 from the cell using the TMB-containing electrolyte is thinner than that on the LiCoO_2 from the cell using the baseline electrolyte. The dominance of the B-O bond on the LiCoO_2 electrode cycled in the TMB-containing electrolyte suggests that TMB participates in the forming of the SEI film and this protection film can effectively suppress the decomposition of the baseline electrolyte^[16].

The peak at 688.0 eV for the C-F bond and the peak at 685.0 eV for the Li-F bond^[11] in the F 1s spectrum both come from the reaction products of the decomposition of LiPF_6 and electrolyte solvents^[32]. In addition, the C-F bond mainly comes from the PVDF. The intensity of the peak for the Li-F bond on the LiCoO_2 cathode cycled in the TMB-containing electrolyte is obviously lower than that on the LiCoO_2 cycled in the baseline electrolyte electrode due to the protection effect of TMB additive. The Li-F bond is usually related to the existence of LiF. Because of the low electric conductivity of LiF,^[33] adding TMB into the electrolyte leads to lower interphase resistance of the LiCoO_2 electrode after 100 cycles. There

is one more peak at 687.2 eV in the F 1s spectrum of the LiCoO₂ electrode in the electrolyte with TMB added. This peak is attributed to the B-F bond^[29], coming from the decomposition product of the TMB additive. The content of the B-F bond takes up a large proportion in the F 1s spectrum of the LiCoO₂ in the electrolyte with TMB added. This suggests that TMB takes part in the film-forming reaction and prevents the severe oxidation of LiPF₆ and electrolyte solvents.

Two weak peaks are observed in the B 1s spectrum of the LiCoO₂ electrode in the electrolyte with the TMB additive. The peak at 192.5 eV is for the B-O band and the peak at 194.0 eV is for the B-F bond^[34], in consistence with the O1s and F1s spectra. The peak at 191.7 eV is assigned to the P 2s in the P-O bond.

The peaks located at 780.4 eV (Co_{2p3/2}) and 795.8 eV (Co_{2p1/2}) appear in the Co 2p spectrum of the three samples^[35] (The Co 2p spectrum of the fresh LiCoO₂ electrode is shown in Figure S2). The appearance of the satellite peaks means the existence of the Co²⁺ ions^[36]. Therefore, the presence of the satellite peaks implies that, after 100 cycles, some Co³⁺ ions in LiCoO₂ are reduced to Co²⁺ ions, probably resulting from the structural degradation of LiCoO₂. Clearly, the satellite peaks of the Co2p are stronger on the LiCoO₂ electrode in the baseline electrolyte. Moreover, the Co 2p_{3/2} peak on the LiCoO₂ electrode in the baseline electrolyte slightly shifts towards the higher binding energy, revealing the existence of Co₃O₄ due to the higher binding energy of Co in Co₃O₄.^[36]

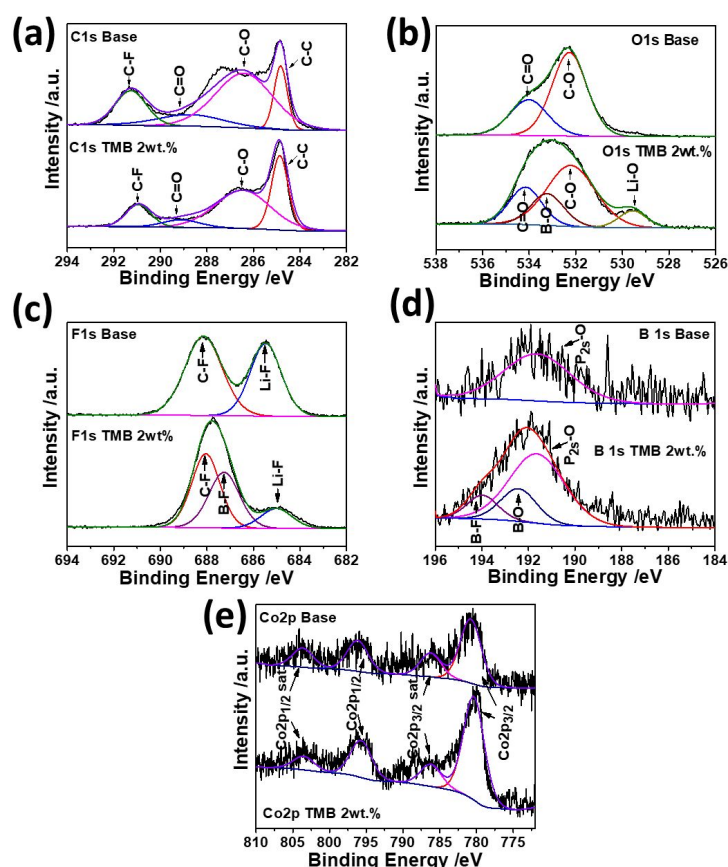


Figure 3. Comparison of the XPS spectra of C 1s (a), O 1s (b), F 1s (c), B 1s (d) and Co 2p (e) of the LiCoO₂ electrodes in the Li/LiCoO₂ cells after 100 cycles using baseline and TMB-containing electrolytes as indicated (The peak at 191.7 eV in the B 1s spectrum is for the P-O bond in the P2s spectrum).

The Li/LiCoO₂ cells using different electrolytes were charged potentiostatically at 4.5V for 10 h initially before the XPS (Figure S3) and FTIR characterization (Figure S4). The FTIR results are

consistent with the outcomes of the XPS spectra of the LiCoO_2 electrodes cycled in different electrolytes after 100 cycles. Although ROCOLi (at 1375 cm^{-1}) is dominant in the two samples according to their FTIR spectra, it can be still observed that absorption peaks characteristic of the B-O (at $1115, 1339\text{ cm}^{-1}$)^[37-38], B-O-B (at 1422 cm^{-1}) and B-F (at 1236 cm^{-1})^[39] bonds, all related to the oxidation of TMB appear on the LiCoO_2 electrode in the TMB-containing electrolyte.

With all above characterization, the possible pathways of the SEI film formed via the TMB oxidation are supposed in Figure 4, explaining the formation of the B-O and B-F bonds in the SEI species. Moreover, according to the Gaussian calculation, the B-O and C-O bonds become longer (about 1.0 pm increased) after one electron is lost, tending to be broken.

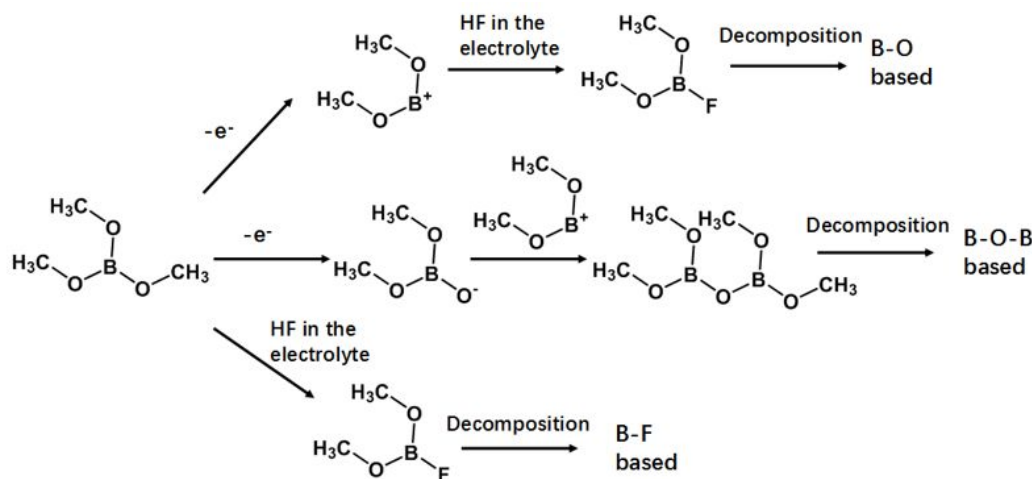


Figure 4. Possible mechanisms for electrochemical oxidative decomposition of TMB

The LiCoO_2 with more than 50% lithium removed is unstable in the organic electrolyte and will lose some of its oxygen. It has been shown by previous calculations and experiments that, when charged to 4.5 V , more than 70% Li will be lost and the oxygen evolution is triggered.^[40-41] Therefore, extraction of too many Li^+ ions will lead to fast capacity fading. The evolved oxygen accumulated at the cathode during the charge process may transform into a superoxide anion radical. This superoxide anion radical attacks the CH_2 moiety of the carbonate solvents, resulting in the electrolyte decomposition and generation of gases such as CO and CO_2 . However, with the existence of TMB additive in the baseline electrolyte, the resulting superoxide anion radical will be entrapped by the electrophilic boron in the SEI layer,^[42] suppressing the further decomposition of the electrolyte.

(4) Protected cathode material

Raman spectroscopy and XRD were applied to appraise the protection effect of the TMB-derived SEI layer on the LiCoO_2 structure. The negligible variation of the XRD patterns of the fresh LiCoO_2 and the LiCoO_2 after cycling 100 times in the TMB-containing electrolyte demonstrate the protection effect of the TMB additive (Figure 5a). In contrast, the (003) and (104) diffractions of the LiCoO_2 in the baseline electrolyte shift towards the lower angles, due to the lattice expansion in the c direction^[43-46]. In addition, the (003) peak of it shows the tendency to split into two peaks, probably due to the irreversible transition from the original hexagonal LiCoO_2 to the cubic spinel (JCDs#42-1467).

The peaks at 482 cm^{-1} and 592 cm^{-1} in the Raman spectrum (Figure 5b) are assigned to the E_g mode (O-Co-O bending) and the A_{1g} mode (stretching vibration of Co-O) in the layered structure of the hexagonal LiCoO_2 , respectively.^[47] Two new peaks appear around 511 cm^{-1} and 674 cm^{-1} after 100 cycles. These two peaks are characteristic of Co_3O_4 , agreeing with the above XPS (Figure 3e) and XRD (Figure

5a) results.^[48] Their presence suggests the degradation of the LiCoO_2 structure and the formation of Co_3O_4 . In addition, the intensity of these two new peaks is stronger for the LiCoO_2 cycling in the baseline electrolyte than for the LiCoO_2 cycling in the electrolyte with the TMB additive. These verify the protection effect of the TMB-derived SEI film on the LiCoO_2 material. Moreover, the position of the two characteristic peaks of LiCoO_2 shift towards the lower wavenumber in the LiCoO_2 in the baseline electrolyte, revealing the weakened Co-O bonding and damaged structure of the LiCoO_2 after cycling.^[49] The existence of Co_3O_4 in the Raman spectra may explain the presence of the Co^{2+} in the Co_{2p} spectrum of the cycled LiCoO_2 (Figure 3).

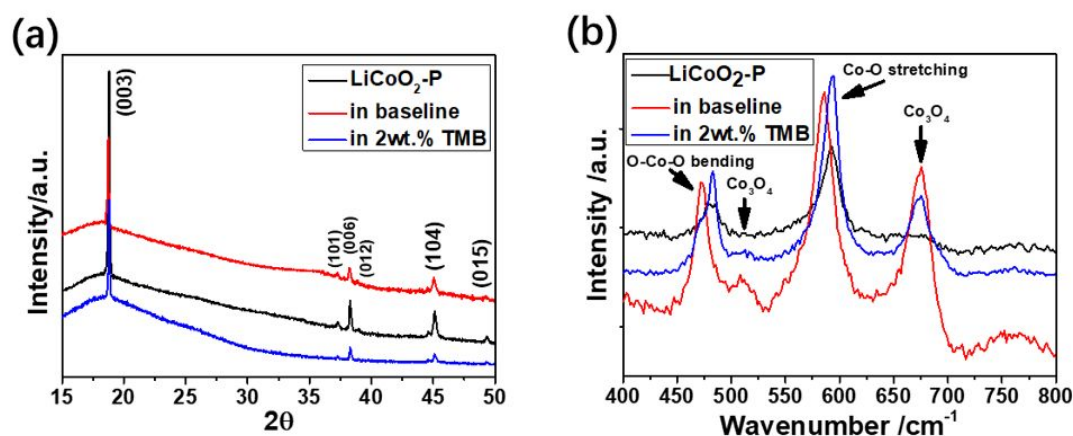


Figure 5. The XRD patterns (a) and the Raman spectra (b) of fresh and 100-cycled LiCoO_2 electrode in the electrolyte with and without TMB additives

Conclusions

Having the cutoff charge voltage increased can boost the energy density of a battery. However, it will also result in fast capacity decay due to severe structural degradation of the cathode materials under high voltages and the deteriorated interfacial instability caused by the continuous oxidation of the electrolyte. In this work, we find that the application of TMB as a film-forming additive to the electrolyte can effectively make the cycling performance of LiCoO_2 at 4.5 V enhanced due to its film-forming capability on the LiCoO_2 . The XPS and FTIR results clearly show that TMB participates in the forming of the SEI layer on the cathode. Characterization of the interfacial composition and morphology after cycling display that TMB decomposes to generate a thin and uniform SEI film that possesses the protection effect on the LiCoO_2 cathode. This film can effectively suppress the mass oxidation of the baseline electrolyte. Moreover, characterization of the structure of the LiCoO_2 before and after cycling in different electrolytes shows that the structure degradation can be mitigated by the compact TMB-derived SEI film. Consequently, the capacity retention of the LiCoO_2 at 0.1 C is improved from 64% to 81% after 100 cycles by simply adding 2.0 wt.% TMB into the baseline electrolyte. Considering its simplicity in operation and effectiveness in performance improvements, TMB has great potential in the research and development of high voltage and high energy-density Li-ion batteries.

Acknowledgments

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Supporting Information

Supporting Information Available: The Supporting Information is available

References

- [1] Zhou, H. M.; Xiao, D. M.; Yin, C. J.; Yang, Z. H.; Xiao, K. W.; Li, J. Enhanced Performance of the Electrolytes Based on Sulfolane and Lithium Difluoro(oxalate)borate with Enhanced Interfacial Stability for $\text{LiNi}_{0.5}\text{Mn}_{1.5}\text{O}_4$ Cathode, *J. Electroanal. Chem.* **2018**, *808*, 293-302
- [2] Hu, M.; Pang, X. L.; Zhou, Z. Recent Progress in High-Voltage Lithium Ion Batteries, *J. Power Sources* **2013**, *237*, 229-242.
- [3] Valanarasu, S.; Chandramohan, R.; Thirumalai, J.; Vijayan, T. A.; Srikumar, S. R.; Mahalingam, T. Effect of Mg Doping on the Properties of Combustion Synthesized LiCoO_2 Powders, *J. Mater. Sci.-Mater. Electron.* **2010**, *21*, 827-832
- [4] Han, B. H.; Paulauskas, T.; Key, B.; Peebles, C.; Park, J. S.; Klie, Robert F.; Vaughey, John T.; Dogan, F. Understanding the Role of Temperature and Cathode Composition on Interface and Bulk: Optimizing Aluminum Oxide Coatings for Li-Ion Cathodes, *ACS Appl. Mater. Interfaces* **2017**, *9*, 14769-14778
- [5] Zheng, Q. F.; Xing, L. D.; Yang, X. R.; Li, X. F.; Ye, C. C.; Wang, K.; Huang, Q. M.; Li, W. S. N-Allyl-N,N-Bis(trimethylsilyl)amine as A Novel Electrolyte Additive to Enhance the Interfacial Stability of A Ni-Rich Electrode for Lithium-ion Batteries, *ACS Appl. Mater. Interfaces* **2018**, *10*, 16843-16851
- [6] Li, Z. D.; Zhang, Y. C.; Xiang, H. F.; Ma, X. H.; Yuan, Q. F.; Wang, Q. S.; Chen, C. H. Trimethyl Phosphite as An Electrolyte Additive for High-Voltage Lithium-Ion Batteries Using Lithium-Rich Layered Oxide Cathode, *J. Power Sources* **2013**, *240*, 471-475
- [7] Abouimrane, A.; Belharouak, I.; Amine, K. Sulfone-Based Electrolytes for High-Voltage Li-Ion Batteries, *Electrochem. Commun.* **2009**, *11*, 1073-1076
- [8] Wang, X. S.; Zheng, X. W.; Liao, Y. H.; Huang, Q. M.; Xing, L. D.; Xu, M. Q.; Li, W. S. Maintaining Structural Integrity of 4.5 V Lithium Cobalt Oxide Cathode with Fumaronitrile as A Novel Electrolyte Additive, *J. Power Sources* **2017**, *338*, 108-116
- [9] Wang, Z. S.; Xing, L. D.; Li, J. H.; Xu, M. Q.; Li, W. S. Triethylborate as An Electrolyte Additive for High Voltage Layered Lithium Nickel Cobalt Manganese Oxide Cathode of Lithium Ion Battery, *J. Power Sources* **2016**, *307*, 587-592
- [10] Zuo, X. X.; Fan, C. J.; Liu, J. S.; Xiao, X.; Wu, J. H.; Nan, J. M. Effect of Tris(trimethylsilyl)borate on the High Voltage Capacity Retention of $\text{LiNi}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{O}_2$ /Graphite Cells, *J. Power Sources* **2013**, *229*, 308-312
- [11] Yu, Q. P.; Chen, Z. T.; Xing, L. D.; Chen, D. R.; Rong, H. B.; Liu, Q. F.; Li, W. S. Enhanced High Voltage Performances of Layered Lithium Nickel Cobalt Manganese Oxide Cathode by Using Trimethylboroxine as Electrolyte Additive, *Electrochim. Acta* **2015**, *176*, 919-925
- [12] Sharabi, R.; Markevich, E.; Fridman, K.; Gershinsky, G.; Salitra, G.; Aurbach, D.; Semrau, G.; Schmidt, M. A.; Schall, N.; Bruenig, C. Electrolyte Solution for the Improved Cycling Performance of LiCoPO_4 /C Composites Cathodes, *Electrochem. Commun.* **2013**, *28*, 20-23
- [13] Freiberg, A.; Metzger, M.; Haering, D.; Bretzke, S.; Puravankara, S.; Nilges, T.; Stinner, C.; Marino, C.; Gasteiger, H. A. Anodic Decomposition of Trimethylboroxine as Additive for High Voltage Li-Ion Batteries, *J. Electrochem. Soc.* **2014**, *161*, A2255-A2261
- [14] Dong, Y. N.; Young, Benjamin T.; Zhang, Y. Z.; Yoon, T.; Heskett, D. R.; Hu, Y. F.; Lucht, B. L. Effect of Lithium Borate Additives on Cathode Film Formation in $\text{LiNi}_{0.5}\text{Mn}_{1.5}\text{O}_4$ /Li Cells, *ACS Appl.*

Mater. Interfaces **2017**, *9*, 20467-20475

[15] Zhu, Y.; Li, Y.; Bettge, M.; Abraham, D. P. Positive Electrode Passivation by LiDFOB Electrolyte Additive in High-Capacity Lithium-Ion Cells, *J. Electrochem. Soc.* **2012**, *159*, A2019-A2117

[16] Wang, Y.; Ming, H.; Qiu, J. Y.; Yu, Z.; Li, M.; Zhang, S. T.; Yang, Y. S. Improving Cycling Performance of LiCoPO₄ Cathode Material by Adding Tris(trimethylsilyl) Borate as Electrolyte Additive, *J. Electroanal. Chem.* **2017**, *802*, 8-14

[17] Cai, Z. J.; Liu, Y. B.; Zhao, J. H.; Li, L.; Zhang, Y. M.; Zhang, J. Tris(trimethylsilyl) Borate as Electrolyte Additive to Improve Performance of Lithium-Ion Batteries, *J. Power Sources* **2012**, *202*, 341-346

[18] Lee, H. S.; Ma, Z. F.; Yang, X. Q.; Sun, X.; McBreen, J. Synthesis of A Series of Fluorinated Boronate Compounds and Their Use as Additives in Lithium Battery Electrolytes, *J. Electrochem. Soc.* **2004**, *151*, A1429-A1435

[19] Lee, H. S.; Yang, X. Q.; Xiang, C. L.; McBreen, J.; Choi, L. S. The Synthesis of A New Family of Boron-based Anion Receptors and the Study of Their Effect on Ion Pair Dissociation and Conductivity of Lithium Salts in Nonaqueous Solutions, *J. Electrochem. Soc.* **1998**, *145*, 2813-2818

[20] Wang, X. S.; Xing, L. D.; Liao, X. L.; Chen, X. F.; Huang, W. N.; Yu, Q. P.; Xu, M. Q.; Huang, Q. M.; Li, W. S. Improving Cyclic Stability of Lithium Cobalt Oxide Based Lithium Ion Battery at High Voltage by Using Trimethylboroxine as An Electrolyte Additive, *Electrochim. Acta* **2015**, *173*, 804-811

[21] Wang, Z. S.; Xing, L. D.; Li, J. H.; Li, B.; Xu, M. Q.; Liao, Y. H.; Li, W. S. Trimethyl Borate as An Electrolyte Additive for High Potential Layered Cathode with Concurrent Improvement of Rate Capability and Cyclic Stability, *Electrochim. Acta* **2015**, *184*, 40-46

[22] Li, J. H.; Xing, L. D.; Chen, J. W.; Zhou, H. B.; Xu, M. Q.; Li, W. S. Improving High Voltage Interfacial and Structural Stability of Layered Lithium-Rich Oxide Cathode by Using A Boracic Electrolyte Additive, *J. Electrochem. Soc.* **2016**, *163*, A2258-A2264

[23] Zhou, A. J.; Wang, W. H.; Liu, Q.; Wang, Y.; Yao, X.; Qing, F. Z.; Li, E. Z.; Yang, T. T.; Zhang, L.; Li, J. Z. Stable, Fast and High-Energy-Density LiCoO₂ Cathode at High Operation Voltage Enabled by Glassy B₂O₃ Modification, *J. Power Sources* **2017**, *362*, 131-139

[24] Li, Y. C.; Wan, S.; Veith, G. M.; Unocic, R. R.; Paranthaman, M. P.; Dai, S.; Sun, X. G.; A Novel Electrolyte Salt Additive for Lithium-Ion Batteries with Voltages Greater than 4.7 V, *Adv. Energy Mater.* **2017**, *7*, 1601397

[25] Wang, C. Y.; Yu, L.; Fan, W. Z.; Liu, J. W.; Ouyang, L. Z.; Yang, L. C.; Zhu, M. Lithium Difluorophosphate as A Promising Electrolyte Lithium Additive for High-Voltage Lithium-Ion Batteries, *ACS Appl. Energy Mater.* **2018**, *1*, 2647-2656

[26] Wang, C. Y.; Yu, L.; Fan, W. Z.; Liu, J. W.; Ouyang, L. Z.; Yang, L. C.; Zhu, M. 3,3' - (Ethylenedioxy)dipropiononitrile as An Electrolyte Additive for 4.5 V LiNi_{1/3}Co_{1/3}Mn_{1/3}O₂/Graphite Cells, *ACS Appl. Mater. Interfaces* **2017**, *9*, 9630-9639

[27] Wang, C. Y.; Yu, L.; Fan, W. Z.; Liu, R.; Liu, J. W.; Ouyang, L. Z.; Yang, L. C.; Zhu, M. Enhanced High-Voltage Cyclability of LiNi_{0.5}Co_{0.2}Mn_{0.3}O₂-Based Pouch Cells via Lithium Difluorophosphate Introducing as Electrolyte Additive, *J. Alloy. Compd.* **2018**, *755*, 1-9

[28] Li, B.; Wang, Y. Q.; Rong, H. B.; Wang, Y. T.; Liu, J. S.; Xing, L. D.; Xu, M. Q.; Li, W. S. A Novel Electrolyte with the Ability to Form A Solid Electrolyte Interface on the Anode and Cathode of A LiMn₂O₄/Graphite Battery, *J. Mater. Chem.* **2013**, *1*, 12954-12961

[29] Zhou, H. M.; Yang, Z. H.; Xiao, D. M.; Xiao, K. W.; Li, J. An Electrolyte to Improve the Deep

- Charge-Discharge Performance of $\text{LiNi}_{0.8}\text{Co}_{0.15}\text{Al}_{0.05}\text{O}_2$ Cathode, *J. Mater. Sci.-Mater. Electron.* **2018**, *29*, 6648-6659
- [30] Matsoso, B. J.; Ranganathan, K.; Mutuma, B. K.; Leretholi, T.; Jones, G.; Coville, N. J. Synthesis and Characterization of Boron Carbon Oxynitride Films with Tunable Composition Using Methane, Boric Acid and Ammonia, *New J. Chem.* **2017**, *41*, 9497-9504
- [31] Quinlan, R. A.; Lu, Y. C.; Kwabi, D.; Shao-Horn, Y.; Mansour, A. N. XPS Investigation of the Electrolyte Induced Stabilization of LiCoO_2 And “ AlPO_4 ”-Coated LiCoO_2 Composite Electrodes, *J. Electrochem. Soc.* **2015**, *163*, A300-A308
- [32] Li, J. H.; Xing, L. D.; Zhang, R. Q.; Chen, M.; Wang, Z. S.; Xu, M. Q.; Li, W. S. Tris(trimethylsilyl)borate as An Electrolyte Additive for Improving Interfacial Stability of High Voltage Layered Lithium-Rich Oxide Cathode/Carbonate-Based Electrolyte, *J. Power Sources* **2015**, *285*, 360-366
- [33] Qin, Z. M.; Hong, B.; Duan, B. Y.; Hong, S.; Chen, Y. C.; Lai, Y. Q.; Feng, J. Tributyl Borate as A Novel Electrolyte Additive to Improve High Voltage Stability of Lithium Cobalt Oxide in Carbonate-Based Electrolyte, *Electrochim. Acta* **2018**, *276*, 412-416
- [34] Liang, F. X.; Yu, J. L.; Chen, J. H.; Wang, D.; Lin, C. D.; Zhu, C. Z.; Wang, M. L.; Dong, L.; Li, C. H. A Novel Boron-Based Ionic Liquid Electrolyte for High Voltage Lithium-Ion Batteries with Outstanding Cycling Stability, *Electrochim. Acta* **2018**, *283*, 111-120
- [35] Lee, S. J.; Han, J. G.; Lee, Y.; Jeong, M. H.; Shin, W. C.; Ue, M.; Choi, N. S. A Bi-Functional Lithium Difluoro(oxalato) Borate Additive for Lithium Cobalt Oxide/Lithium Nickel Manganese Cobalt Oxide Cathodes and Silicon/Graphite Anodes in Lithium-Ion Batteries at Elevated Temperatures, *Electrochim. Acta* **2014**, *137*, 1-8
- [36] Zhang, B. T.; Zhang, Y.; Teng, Y. G. Electrospun Magnetic Cobalt-Carbon Nanofiber Composites with Axis-Sheath Structure for Efficient Peroxymonosulfate Activation, *ACS Appl. Surf. Sci.* **2018**, *452*, 443-450
- [37] Nakahara, H.; Nutt, S. Compounds in Solid Electrolyte Interface (SEI) on Carbonaceous Material Charged in Siloxane-Based Electrolyte, *J. Power Sources* **2006**, *160*, 1355-1360
- [38] Peak, D.; Luther III, G. W.; Sparks, D. L. ATR-FTIR Spectroscopic Studies of Boron Acid Adsorption on Hydrated Ferric Oxide, *Geochim. Cosmochim. Acta* **2003**, *67*, 2551-2560
- [39] Sun, B. J.; Jin, Q.; Tan, L. S.; Wu, P. Y.; Yan, F. Trace of the Interesting “V”-Shaped Dynamic Mechanism of Interactions Between Water and Ionic Liquids, *J. Phys. Chem. B* **2008**, *112*, 14251-14259
- [40] Kim, Y.; Jeong, D. Y.; Han, S. C. First-Principles Investigation of the Gas Evolution from the Cathodes of Lithium-Ion Batteries during the Storage Test, *J. Mater. Sci.* **2014**, *49*, 8444-8448
- [41] Shen, B.; Zuo, P. J.; Fan, P.; Yang, J.; Yin, G. P.; Ma, Y. L.; Cheng, X. Q.; Du, C. Y.; Gao, Y. Z. Improved Electrochemical Performance of NaAlO_2 -Coated LiCoO_2 for Lithium-Ion Batteries, *J. Solid State Electrochem.* **2017**, *21*, 1195-1201
- [42] Han, J. G.; Bin Lee, J.; Cha, A.; Lee, T. K.; Cho, W.; Chae, S.; Kang, S. J.; Kwak, S. K.; Cho, J.; Hong, S. Y.; Choi, N. S. Unsymmetrical Fluorinated Malonatoborate as An Amphoteric Additive for High-Energy-Density Lithium-Ion Batteries, *Energ. Environ. Sci.* **2018**, *11*, 1552-1562
- [43] Ubbelohde, A. R. Intercalation into Carbons with Varying Degree of Graphitization, *Mater. Sci. Eng.* **1997**, *31*, 341-350
- [44] Ohzuku, T.; Ueda, A. Solid-State Redox Reactions of LiCoO_2 (R3m) for 4 Volt Secondary Lithium Cells, *J. Electrochem. Soc.* **1994**, *141*, 2972-2977

- [45] Amatucci, G. G.; Tarascon, J. M.; Klein, L. C. CoO_2 , the End Member of the Li_xCoO_2 Solid Solution, *J. Electrochem. Soc.* **1996**, *143*, 1114-1123
- [46] Chen, Z. H.; Lu, Z. H.; Dahn, J.R. Staging Phase Transitions in Li_xCoO_2 , *J. Electrochem. Soc.* **2002**, *149*, A1604-A1609
- [47] Mendoza, L.; Baddour-Hadjean, R.; Cassir, M.; Pereira-Ramos, J. P. Raman Evidence of the Formation of LT- LiCoO_2 Thin Layers on NiO in Molten Carbonate at 650 Degrees C, *ACS Appl. Surf. Sci.* **2004**, *225*, 356-361
- [48] Venkatesh, R.; Dhas, C. Ravi; Sivakumar, R.; Dhandayuthapani, T.; Sudhagar, P.; Sanjeeviraja, C.; Raj, A. Moses Ezhil Analysis of Optical Dispersion Parameters and Electrochromic Properties of Manganese-Doped Co_3O_4 Dendrite Structured Thin Films, *J. Phys. Chem. Solids* **2018**, *122*, 118-129
- [49] Ozawa, Y; Yazami, R; Fultz, B Self-Discharge Study of LiCoO_2 Cathode Materials, *J. Power Sources* **2003**, *119-121*, 918-923

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